

Head of Software Engineering & Solution Architecture

Technical Exercise | Candidate Brief

YOUR CHALLENGE

Conceive, build and deploy a new digital experience that meaningfully supports Alpha's mission, participants, churches, teams or global operations. You choose the problem, user, platform, product form, stack and tools.

TIMEBOX

- You have a 48-hour submission window.
- Maximum expected effort: 12 hours.
- Your start time and deadline will be supplied separately.
- At cutoff, all submitted artifacts are frozen. Make no further code, content, configuration, deployment, documentation or bug-fix changes unless requested by the panel.

MINIMUM PRODUCT

- Fully deployed and accessible to assessors; local-only submissions are not accepted.
- Working authentication with dedicated evaluator and administrator accounts.
- Persistent data and at least one complete end-to-end workflow.
- A mobile-responsive web experience, or a distributable native mobile build.
- A credible, presentable experience aligned with Alpha.

SUBMISSION BUNDLE

- **Live product:** deployment URL and dedicated evaluator and administrator credentials.
- **Source:** read access to the Git repository, the submitted commit SHA or tag, and a ZIP snapshot excluding dependencies and secrets.
- **Walkthrough:** a five-minute video covering the product and important decisions.
- **Executive summary:** a one-page PDF describing the user and problem, solution, architecture, decisions and trade-offs, security, testing, limitations and next steps.
- **Product and technical dossier:** a comprehensive account of the submission, covering every area listed below.
- **Complete process archive:** all evidence of your planning and work - every exercise-related artifact and work record from start to cutoff, including planning, research, notes, sketches, task breakdowns, decision records, drafts, experiments, iterations, test evidence, working logs and work-session histories. Preserve original sequence and timestamps where available. Redact only secrets or third-party data you are not authorized to share, and identify any redactions.
- **Engineering manifesto:** a separate, easy-to-read one-page PDF setting out your vision for the future of engineering at Alpha and what you would bring to it.
- **README:** setup, deployment, service inventory and reproduction instructions.

COMPREHENSIVE DOSSIER: REQUIRED COVERAGE

- **Product and experience:** purpose, problem, users, user stories, journeys, requirements, scope and key experience decisions.
- **System design:** clear explanations plus architecture, context, component and deployment diagrams; complete database schemas and data model; data flows; APIs and contracts; every integration, dependency and system connection; authentication and access; infrastructure and environments.
- **Quality and security:** testing, observability, security model, threat and risk analysis, data and privacy decisions, and known limitations.
- **Future and production:** roadmap, product and technical vision, remaining work, and the time, effort, people, roles, costs, dependencies and assumptions needed to take the product to production.

Submission conditions: Keep the deployment available through the interview and debrief. • Be ready to demonstrate the product and defend the submission. • All submitted materials must correspond to the version frozen at cutoff.

Access and data hygiene: use only data and credentials you are authorized to share. Do not include production, personal, billing or reusable credentials. Keep secrets out of source control and archives.